EPSR-test-ex52 coord: coordination-number distributions

Program(s): coord | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `coord`, which for each distribution counts the `atom2` atoms within [rmin, rmax] of every `atom1` atom and histograms these coordination numbers, together with the mean bond length for each coordination number [1]. Optionally only `atom1` atoms in a window `rminrmaxc` from a centre type are used, measured in 3D (`atom-c`), along z (`atom-z`) or in the xy-plane (`atom-r`).

Code exercised

Fortran (original, fortran_src/):

- `coord.f`: `run_coord` (keywords, `fort.14` means), `find_coord` (neighbour counting, centre-atom windows)
- `read_and_save_files.f`: `read_input_file`, `initgofz`, `outgofz` (histogram file, running average and spread), `save_files_others`; `auxiliary_routines.f`: `r_int`
- `setup_symbols_generic.f`: `setup_symbols_coord`

Rust (port, rust/epsr-core/src/):

- `tools/coord.rs`: `run`, `find_coord`
- `tools/anal_common.rs`: `match_centre`, `outgofz`, `write_fort14`
- `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `COORD`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å³), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [2]. Coordination shells: water-water hydrogen bonds O1···H1 1.4-2.4 Å, water O1 around S 2.5-4.5 Å and DMSO O···water H1 1.2-2.5 Å (`hbond`); O1-O1 2.0-3.4 Å for O1 within 1 Å in z of an O1 (`slabz`); O1-O 1.0-3.5 Å within 3 Å of S in the xy-plane (`cylr`); O1-O1 and O1-H1 coordination of water in the shells 3-5 Å and 2.5-4 Å from S (`centreS`).

Inputs

Four keyword files `hbond`, `slabz`, `cylr`, `centreS` `.COORD.dat` (3, 1, 1 and 2 distributions; `slabz` with `nsumt -1`, the others `nsumt 0`); `200DMSO400Water.ato` copied unchanged from EPSR-test.

How the accuracy was verified

`commands.txt` (4 program calls, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.

The four commands run in sequence, the list twice, so three inputs accumulate over the two passes. Compared: the four `.COORD.n01` files (histogram and mean bond length columns), the four rewritten `.dat` files and `fort.14` of the last run (`centreS`), whose line uses Fortran format reversion; the reference gives mean coordination numbers 2.593 (rms 1.057) for O1-O1 and 1.058 (rms 0.628) for O1-H1 around S. The README states that `fort.14` of `hbond` (`ltypc = 0` branch) was also checked identical in a separate single-command run. The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).

[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.